A061 — Nuclear Clock / Swirl Clock Closure Falsifier Version v0.1.1 (SST Framework v1.0.7)

SST-Workbench: independent falsification protocol

8 October 2026

1 Identification and epistemic status

Independent research-family A061, scientific revision v0.1.1. This release migrates the v0.1.0 study to the immutable SST Falsifier Framework 1.0.7. It does not establish SST clock physics. The strict Python FP64 reference and C++17/pybind FP64 certification lanes are distinct. SYCL is disabled in the CPU profile.

2 Research question

Can the local $S_t^{\circlearrowleft}$ factor proposed in SST-28 predict a species-dependent frequency shift in measured ^{229}Th/Yb clock comparisons, distinct from known environmental frequency shifts and laser-feedback transfer functions? A universal clock rescaling alone cannot change a frequency ratio.

3 Null and alternatives

H_0: observed differential frequency excursions arise from species-independent time scaling cancellation, transitions' intrinsic energy shifts, thermal/crystalline effects, and feedback dynamics; no independently committed SST transfer law is available. H_1: an independently derived SST susceptibility predicts, before observing target data, nonzero signed differential changes and uncertainty bounded by the measured data. Neither mathematical compatibility nor a synthetic reconstruction is empirical support for H_1.

4 Assumptions and edge cases

SST-28 is treated as an explicitly defined, dimensionless geometric ansatz. The $\omega = 0$ locus does not imply $v_\varphi = 0$. A singular circulation model is never substituted for the smooth finite-core profile without an explicit regularization. The two independent nuclear laboratories are not assumed phase-synchronous. Small/large radius checks: $S(0) = S(\infty) = 1$, while at $r = r_c$ the smooth reference has $S = \sqrt{1 - e^{-2}}$. Missing time series or native dependencies produce `UNRESOLVED`; the harness must not synthesize experimental data.

5 Symbols and units

Symbol	Definition	SI unit	Dimensions
x	r/r_c	1	1
S	SST clock factor	1	1
f	Nuclear/atomic clock frequency	Hz	T^{-1}
ω_z	Axial vorticity	s^{-1}	T^{-1}
t	Chronological timestamp	s	T

P	Normalized periodogram power	1	1

6 Equations of the testable models

Define $x = r/r_c$ and use the smooth reference profile $v_\varphi/\|\mathbf{v}_\circlearrowleft\| = xe^{-x}$:

$$S(x) = \sqrt{1 - x^2 e^{-2x}}, \qquad \tilde{\omega}_z(x) = e^{-x}(2 - x). \tag{1}$$

The cited circulation-pole identification $\kappa = 2\pi r_c \|\mathbf{v}_\circlearrowleft\|$ gives

$$S_n(x) = \sqrt{1 - n^2/x^2}; \quad S_{n=1}(x = 1) = 0. \tag{2}$$

SST-28 also prints approximately 0.9603 at that same point. This is an algebraic inconsistency subject to gate G3; it is *not* silently repaired. Likewise, $\omega_z(2r_c) = 0$ while $S(2r_c) < 1$. For a clock transition with an independently defined local Hamiltonian gap:

$$f_i^{\mathrm{obs}} = S_i \, \Delta E_i^{\mathrm{loc}}/h, \qquad \delta \ln f_i = \delta \ln S_i + \delta \ln \Delta E_i^{\mathrm{loc}}. \tag{3}$$

Both sides have dimensions T^{-1}; the two logarithmic variations are dimensionless. When $S_{\mathrm{Th}} = S_{\mathrm{Yb}}$ it cancels from $f_{\mathrm{Th}}/f_{\mathrm{Yb}}$.

7 Algorithm and computational state

Source-pinned published-summary reconstruction; analytic SST-28 checks; structural identifiability; a delayed proportional error-feedback synthetic null; optional generalized least-squares sinusoidal scan with a global block-resampled peak threshold; independent pybind11 C++/NumPy parity; no cross-laboratory pseudo-replication. A measured periodic residual would only be a candidate instrumental/physical anomaly, never SST confirmation without a frozen transfer law. The C++ pass is a numerical-kernel pass, not a clock experiment.

8 Sources and provenance

Primary experimental PDFs are bundled (arXiv:2606.04997v2 and arXiv:2606.08870v2). Huang and Toscani are independent laboratories, but their data are not treated as contemporaneous co-phased time series. SST-28 is the explicit prior theory. The experimental data repositories (Zenodo 22657608 and Figshare 33386191) are external optional inputs, never auto-invented. Public summary values and synthetic nulls have distinct evidence classes. The framework records actual source paths and file digests.

9 Independent gate plan

G0: protocol/source integrity; G1: published frequency reconstruction; G2: universal-factor cancellation; G3: SST-28 algebra; G4: crystal nuisance identifiability; G5: strict C++ FP64 parity; G6: optional GPU; G7: optional multilibrary; G8: physical transfer closure; G9: reveal/deferred; G10: feedback-only null; G11: raw time-series periodogram; G12: theory prediction; G13: cross-lab replication. Gates have no scientific predecessor dependencies so the G3 failure does not stop G4–G13; framework integrity checks remain fatal.

10 Numerics and convergence

FP64 Python is the authoritative profile reference. The native C++ gate is a strict, separately compiled double-precision pybind kernel, controlled through framework build provenance. Tolerance for the dimensionless profile: 10^{-11}. Monte Carlo (if raw timing is valid) uses 160 block-resampled trials and 240 geometric frequencies, with trial-corrected global threshold 5%; this exploratory test is not yet a calibrated clock-feedback likelihood.

11 Backend qualification

Framework v1.0.7 verifies the protocol before importing the scientific pipeline. Compiler/Python ABI details are recorded by the native builder; a missing compiler or pybind11 yields `UNRESOLVED` instead of silent Python fallback. The native source is included; no precompiled Windows binary is published. GPU lanes are `NOT_APPLICABLE` under CPU mode.

12 Blindness and withholding rules

The frozen public science contract and the private reveal payload have nonce-backed commitments. Synthetic cases remain anonymous in all public outputs, and are expressly not physics evidence. Reveal requires preregistered G12=PASS in FULL/CERTIFY with no unresolved gates. A routine `REVEAL_IF_ALLOWED` therefore withholds reveal while the SST transfer remains unspecified. The blind-source scan excludes third-party runtime/vendor ownership but examines instance-owned source.

13 Statistical discrimination

Reported frequency differences, standard deviations, Allan instabilities, material shifts, in-loop servo response and global search false-alarm rates must remain separate quantities. Clock and crystal systematic uncertainty cannot be interpreted as a universal local-time effect. An observed peak in a single periodogram is not a cross-laboratory replication. No fitted parameter is allowed to be called an independently predicted SST coupling.

14 Automatic run results

Gate	Status	Meaning
G0	PASS	Framework protocol verified upstream; source files verified locally.
G1	PASS	Huang published two-crystal shift algebra; stability is not systematic accuracy.
G2	PASS	Universal factor cancellation identity, not SST empirical support.
G3	FAIL	SST-28 Eq. 23/24 disagree under printed kappa=2*pi*rc*vchar; omega(2rc)=0 does not imply S(2rc)=1.
G4	UNRESOLVED	Two crystals do not identify unknown SST clock susceptibility separately from local crystal/temperature shifts.
G5	PASS	Strict native pybind11 FP64 profile parity via frozen framework C++ build system.
G6	NOT_APPLICABLE	Disabled in CPU profile; optional SYCL/multilibrary lanes not promoted to precision certification.
G7	NOT_APPLICABLE	Disabled in CPU profile; optional SYCL/multilibrary lanes not promoted to precision certification.
G8	UNRESOLVED	No independently derived SST constitutive map (Th/Yb or crystal-specific), hence no empirical Swirl Clock closure.
G10	PASS	Instrument-only delayed feedback generates a structured spectrum without SST; synthetic control.
G11	UNRESOLVED	Raw Toscani cadence required and only evaluated in FULL/CERTIFY; summary statistics cannot replace it.
G12	UNRESOLVED	SST species-resolved differential transfer coefficient not independently precommitted; local payload cannot be assumed predic

Gate	Status	Meaning
G13	UNRESOLVED	Two independent clock experiments do not establish a synchronous shared-phase spectral replication.
G9	DEFERRED	Separate reveal-only phase; all scientific conclusions remain blind.

15 Interpretation boundaries

G2=PASS would be a cancellation identity, not positive evidence for SST. G3=FAIL indicates an internally inconsistent source expression. G5=PASS would qualify numerical parity only. Missing raw data, absent a differential SST susceptibility, or an unmatched timebase prevents any claim about nuclear time dilation. Failing one independent scientific gate does not stop later exploratory gates.

16 Replication and bibliography

Run `run_all.cmd SELFTEST`, `run_all.cmd FREEZE`, `run_all.cmd BASIC`, `run_all.cmd BUILD_NATIVE`, `run_all.cmd FETCH`, `run_all.cmd FULL`, then `run_all.cmd REVEAL_IF_ALLOWED`. The unmodified canonical framework resides separately in `06_templates`.

References

[1] O. Iskandarani (2026), "Time from Swirl: A Hydrodynamic Origin of Proper Time," SST-28, Zenodo, doi:10.5281/zenodo.17877012.

[2] L. Toscani De Col et al. (2026), "A thorium-229 optical nuclear clock with feedback loop," arXiv:2606.04997v2, `https://arxiv.org/abs/2606.04997`.

[3] B. Huang et al. (2026), "A nuclear clock synchronized to 229Th," arXiv:2606.08870v2, `https://arxiv.org/abs/2606.08870`.

[4] J. D. Scargle (1982), "Studies in astronomical time series analysis. II," *Astrophysical Journal* **263**, 835, doi:10.1086/160554.